

Dana Martinez Gonzalez

(559) 630-2682 | danapaolawork17@gmail.com | Portfolio: danapaolawork17-creator.github.io

PROFILE

Mechanical Engineering student with hands-on experience in design and aerospace, skilled in CAD and project coordination, with a strong interest in sustainable engineering solutions.

EDUCATION

California State University, Fresno – Lyles College of Engineering

Bachelor of Science in Mechanical Engineering | Minor in Aerospace Engineering

Master of Science in Mechanical Engineering (Finish in Five Program)

Expected Graduation: Spring 2027 (BS) | Spring 2028 (MS) | GPA: 3.52

WORK EXPERIENCE & RESEARCH

NASA Pathways Student Trainee, NASA Armstrong Flight Research Center, Edwards, CA

August 2026 – December 2026

- Supporting X-59 QueSST structural analysis and stress evaluations within the Aerostructures Branch.
- Monitoring real-time flight test data in the control room to identify anomalies, gaining flight controls experience.
- Collaborating with Lockheed Martin engineers to analyze structural and flight test data.
- Writing Python scripts to process flight test and structural data more efficiently.
- Modeling structural components in SolidWorks and PTC Creo for design reviews.

Test Engineering Intern, NASA Armstrong Flight Research Center, Edwards, CA

June 2026 – August 2026

- Developed a quantum sensor prototype integrating hardware and software components, including RF, optics, and PCB-related system design and testing.
- Performed thermal testing using a Thermotron environmental chamber to characterize sensor performance across operating temperature ranges.
- Conducted vibration analysis and testing to evaluate sensor performance under flight-representative dynamic loading.
- Designed and implemented TEC (thermoelectric cooler) controller logic to regulate and stabilize laser temperature for the sensor's optical subsystem.
- Supported mechanical design and system integration using CAD and experimental validation, collaborating with a multidisciplinary team to refine system performance.

Rotorcraft Aeromechanics Intern, NASA Ames Research Center, Moffett Field, CA

June 2025 – August 2025

- Developed a Python based rotor power calculator with a custom GUI using interpolated CAMRAD II power coefficient tables to evaluate rotorcraft performance.
- Automated airfoil slicing from 3D rotor blade geometry in Rhino using rhinoscriptsyntax to support aerodynamic analysis workflows.
- Supported research on next generation vertical flight vehicles by applying CFD, aeroelastic analysis, and wind tunnel testing to assess performance, noise, vibration, and ride quality.

Engineering Intern, Duncan Enterprises (iLoveToCreate), Fresno, CA

December 2023 – May 2025

- Created packaging specifications and engineering drawings for new product lines using SolidWorks and ArtiosCAD.
- Conducted time studies and data analysis to support production improvements and efficiency reporting.
- Wrote standardized packaging instructions and collaborated with Operations and Quality leadership.

PROJECTS & LEADERSHIP

Chassis and Propulsion Team Member, MATE ROV Competition, California State University, Fresno,

November 2025 – Present

- Designed and evaluated a six thruster propulsion layout to enable full 6 DOF control and station keeping in flowing water.
- Contributed to chassis layout, thruster placement, and shroud considerations to improve stability, maneuverability, and safety; developed tether routing and strain relief concepts to protect onboard electronics.

President, Society of Women Engineers (SWE), California State University, Fresno,

June 2026 – May 2027

- Lead a team of seven chapter officers, overseeing programming, professional development events, and K-14 outreach initiatives.
- Direct the chapter's WE26 national conference sponsorship campaign, managing corporate outreach, a tiered sponsorship packet, and officer task assignments to fund student travel to Boston.

TECHNICAL SKILLS

Engineering & Analysis: Systems Engineering | CFD | FEA | Numerical Methods | Engineering Testing (MTS) | Environmental Testing (Thermotron chambers) | Controls

Software & Tools: SolidWorks | PTC Creo | Siemens NX | ANSYS | MATLAB | Simulink | Python | AutoCAD | Revit | SketchUp | ArtiosCAD

Hardware & Systems: RF Systems | PCB Design | Arduino | Soldering | TEC Controllers | Data Acquisition | Instrumentation

Programming: Python | MATLAB | C++

Manufacturing: CNC Machining | Manual Machining